

Chapter: Dynamic Programming

LeetCode 813 - Largest Sum of Averages

Intuition & Notes

- Once again a **classical partition DP**.
- *"kabhi bhi splitting array into multiple parts dikhe, ek baar dp[i][k] zaroor sochna maza aayega"*
- Here, dp[i][k] means the first i elements split into k parts.
- It represents the maximum sum of averages of each subarray.

C++ Implementation

```
class Solution {
public:
    double largestSumOfAverages(vector<int>& nums, int k) {

        //once again a classical partition dp
        //kabhi bhi splitting array into multiple parts dikhe
        //ek baar dp[i][k] zaroor sochna maza aayega
        //here dp[i][k] means first i elements k parts
        //represents max sum of averages of each subarray

        int n = nums.size();
        vector<double> pref(n + 1, 0);

        for(int i = 0; i < n; i++) {
            pref[i + 1] = pref[i] + nums[i];
        }

        vector<vector<double>> dp(n + 1, vector<double>(k + 1, -DBL_MAX));
```

```
dp[0][0] = 0;

for(int i = 1; i <= n; i++) {
    for(int parts = 1; parts <= k; parts++) {

        for(int j = parts - 1; j < i; j++) {

            double lastavg = (pref[i] - pref[j]) / (i - j);
            double curr = dp[j][parts - 1] + lastavg;

            dp[i][parts] = max(dp[i][parts], curr);
        }
    }
}

return dp[n][k];
}
};
```